$$y[n] = \frac{5}{6} y[n-1] - \frac{1}{6} y[n-2] + x[n] \quad \text{— (1)}$$

$$x[n] = \delta[n] - \frac{1}{3}\delta[n-1]$$

Take z – tran of (1)

$$\Rightarrow Y(z) = \frac{5}{6} z^{-1} Y(z) - \frac{1}{6} z^{-2} Y(z) + X(z)$$

$$\Rightarrow \frac{Y(z)}{X(z)} = \frac{1}{1 - \frac{5}{6} z^{-1} + \frac{1}{6} z^{-2}} = H(z)$$

↳ Transfer funcⁿ

$$\Rightarrow H(z) = \frac{1}{\left(1 - \frac{1}{2} z^{-1}\right)\left(1 - \frac{1}{3} z^{-1}\right)}$$

Now transform the input

$$\Rightarrow X(z) = 1 - \frac{1}{3} z^{-1}$$

$$\because Y(z) = H(z) \cdot X(z)$$

$$\Rightarrow Y(z) = \frac{1}{1 - \frac{1}{2} z^{-1}}$$

$$\Rightarrow y[n] = \left(\frac{1}{2}\right)^n u[n]$$

For stability

$$\sum_0^\infty \left(\frac{1}{2}\right)^n \quad \text{should be finite}$$

$$\Rightarrow \sum_0^\infty \left(\frac{1}{2}\right)^n = 2 \rightarrow \text{finite} \Rightarrow \text{Hence, stable}$$

For casualidty, the response must not exist before input

$\because y[n] = \left(\frac{1}{2}\right)^n u[n]$

$\Rightarrow y[n] = 0 \quad \forall \; n < 0$

$\Rightarrow$ Casual